Medha Agarwal

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Education

Dec 2026	**PhD, Statistics** University of Washington Advisor: Alex Luedtke
May 2021	**Bachelor of Science, Mathematics and Scientific Computing** Indian Institute of Technology Kanpur Advisor: Dootika Vats | Minor: English Literature

Research Interests

Causal inference and semiparametric efficiency theory; optimal transport and distributional inference; uncertainty quantification and conformal prediction, particularly under distribution shift; gradient flows and mean-field dynamics of learning algorithms; machine learning for scientific applications, particularly ecology.

Peer-Reviewed Publications

[1] **Medha Agarwal** and A. Luedtke. Sinkhorn treatment effects: A causal optimal transport measure. In *Forty-third International Conference on Machine Learning (ICML)*, 2026

[2] **Medha Agarwal**, K. Rafiq, R. Mehta, B. Abrahms, and Z. Harchaoui. Leveraging machine learning and accelerometry to classify animal behaviours with uncertainty. *Methods in Ecology and Evolution*, 17(2):567–584, 2026

[3] **Medha Agarwal**, Z. Harchaoui, G. Mulcahy, and S. Pal. Langevin diffusion approximation to same marginal Schrödinger bridge. *Journal of Functional Analysis*, page 111494, 2026

[4] **Medha Agarwal** and J. Xu. Quasi-Newton Acceleration of EM and MM Algorithms via Broyden's Method. *Journal of Computational and Graphical Statistics*, 2023

[5] **Medha Agarwal**, D. Vats, and V. Elvira. A principled stopping rule for importance sampling. *Electronic Journal of Statistics*, 2022

[6] **Medha Agarwal** and D. Vats. Globally centered autocovariances in MCMC. *Journal of Computational and Graphical Statistics*, 2022

Preprints

[7] **Medha Agarwal**, Z. Harchaoui, G. Mulcahy, and S. Pal. Iterated Schrödinger bridge approximation to Wasserstein gradient flows. *arXiv preprint arXiv:2406.10823*, 2024

Research Positions

Jun-Dec 2025	**Abrahms Lab**, Research Intern, PI: Briana Abrahms, *Seattle, WA*
Mar-Jun 2024	**eScience Institute**, Research Assistant, PI: Zaid Harchaoui & Briana Abrahms, *Seattle, WA*
2023	**Institute for Foundations of Data Science**, Research Assistant, PI: Zaid Harchaoui, *Seattle, WA*
Jul-Sept 2020	**University of Edinburgh**, Research Intern, PI: Victor Elvira, *Edinburgh, Scotland*
May-Jul 2020	**Duke University**, Research Intern, PI: Jason Xu, *Durham, NC*

Honors and Awards

2026	2026 IMS International Conference on Statistics and Data Science, Student Travel Award *(declined)*
2025	Graduate Student Conference Presentation Award, University of Washington
2024	Winner, Student Paper Competition, ASA Conference on Statistical Learning and Data Science
2023	Hannan Graduate Student Award, Institute of Mathematical Statistics
2022	Center for Statistics and the Social Sciences Travel Award, University of Washington

2022	Scholarship recipient, Institute for Foundations of Data Science *Supported by the NSF Transdisciplinary Research in Principles of Data Science (TRIPODS) program*
2021	Department Fellowship, UW Statistics
2021	Proficiency Medal, Department of Mathematics and Statistics, IIT Kanpur *Awarded for the best academic performance among graduating students in each department*
2017	Academic Excellence Award - Dr. Sangeeta Goel Memorial Award, IIT Kanpur *Awarded for the highest womens' All India Rank (365) in IIT-Joint Entrance Exam*

Research Talks & Presentations

⋆ Invited talk • Oral presentation ◇ Poster

Sinkhorn Treatment Effects: A Causal Optimal Transport Measure

[1] Nov 2026	ASA Section on Statistical Learning and Data Science Conference ⋆	*Brooklyn, NY*
[2] Oct 2026	Department of Health Care Policy, Harvard Medical School ⋆	*Boston, MA*
[3] Jul 2026	International Conference on Machine Learning (ICML) ◇	*Seoul, South Korea*
[4] May 2026	Optimal Transport + Optimization Workshop, NCCR SwissMAP •	*Les Diablerets, Switzerland*

Iterated Schrödinger Bridge Approximation to Heat Flow

[5] Aug 2025	Joint Statistical Meetings • *Advances in Generative Models Session*	*Nashville, TN*
[6] Feb 2025	Institute for Foundations of Machine Learning • *Mathematics of Deep Learning Workshop*	*Austin, TX*
[7] Jan 2025	Joint Mathematics Meetings ⋆ *Mathematics of Adversarial, Interpretable, and Explainable AI*	*Seattle, WA*
[8] Nov 2024	ASA Conference on Statistical Learning and Data Science ◇	*Newport Beach, CA*
[9] Oct 2024	SIAM Mathematics of Data Science ◇	*Atlanta, GA*

Leveraging Machine Learning and Accelerometry to Classify Animal Behaviors

[10] Oct 2025	Institute for Foundations of Data Science Seminar •	*Seattle, WA*
[11] Feb 2025	UW Data Science Seminar ⋆ *Seminar Series for AI@UW Seed Grant Awardees*	*Seattle, WA*

Globally centered autocovariances in MCMC

[12] May 2023	BayesComp ⋆ *Session on Measuring Quality of MCMC Samples*	*Levi, Finland*

Selected Workshops Attended

† Invited attendee

Jul 2025	**Wasserstein Gradient Flows in Math and ML Workshop** † *Banff International Research Station*	*Banff, Canada*
Aug 2022	**Deep Learning Theory Workshop and Summer School** *Simons Institute for the Theory of Computing*	*Berkeley, CA*
Jul 2020	**noRth 2020: A Virtual Conference for R Users**	*Minneapolis, MN*

Teaching Experience

Spring 2026	STAT 498/591 - Special Topics: Optimization and Sampling
Winter 2026	STAT 311 - Elements of Statistical Methods
Spring 2025	CSE 546 - Machine Learning for Big Data
Autumn 2024	STAT 512 - Statistical Inference I
Winter 2024, 2023	STAT 513 - Statistical Inference II
2021–2022	STAT 311 - Elements of Statistical Methods

Industry Experience

Summer 2026	**Quantbot Technologies** Quantitative Researcher Intern	*New York City, NY*
	• Developing statistical learning methods for equity return prediction, with an emphasis on modeling idiosyncratic stock movements.	
Summer 2024	**Amazon Science** Applied Scientist Intern	*New York City, NY*
	• Developed multi-horizon probabilistic forecasting methods for multivariate time series using MQ-CNN; achieved $> 95\%$ empirical coverage on 5+ years of production-scale demand data.	
Summer 2023	**Amazon Science** Applied Scientist Intern	*Bellevue, WA*
	• Fine-tuned a 40B foundation model using reinforcement learning from human feedback, integrating human preferences with product-description corpora. • Project paper featured in Amazon Machine Learning Conference 2023.	

Professional Service

Reviewer, *Journal of Statistical Planning and Inference*, 2026

Academic Service

2022-23	Leader, Normalizing Flows Working Group (Webpage)	*UW Seattle*
Winter 2021	Statistics Undergraduate Directed Reading Program (SPA-DRP) Mentor	*UW Seattle*
Autumn 2021	Pre-application Review Service (PARS) volunteer	*UW Seattle*
2019-20	Student Undergraduate Committee Student Nominee	*IIT Kanpur*